

THEORY AND METHODS

TESTING FOR OUTLIERS AND INFLUENTIAL
OBSERVATIONS IN MULTIPLE REGRESSION
USING RESTRICTED LEAST SQUARES

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Summary: When certain prior information is known on the parameters β of the linear model $Y = X\beta + \epsilon$ then there can be a considerable improvement in the estimation of these parameters. In this paper, the prior information is in the form of linear restrictions on the parameters. Test statistics and diagnostics are developed to test for outliers and to detect influential observations when the residuals are analyzed, after these residuals have been computed using the restrictions on the parameters. The distributional results for both single and multiple outliers and influential observations are given. The distributions of the test statistics and diagnostics are complicated non-central distributions. Under certain assumptions (for example exact restrictions) the distributions simplify to central distributions. Comparisons can then be made with traditional test statistics and diagnostics. The results are demonstrated using a simulated data set and also two known practical examples in the literature. The data used are those of Belsley, Kuh and Welsch (1980) and Chatterjee and Hadi (1986).

1. Introduction

In many applications in statistics, econometrics and finance, the analyst may have prior information on particular regression coefficients or linear

combinations thereof, and may wish to incorporate this information into the model. Alternatively, the prior information may form constraints upon the regression coefficients. In this paper we investigate what such constraints or restrictions (RLS) can have on analyzing the computed residuals or functions of the computed residuals.

2. Analyzing RLS Residuals

Assume the linear model,

$$Y = X\beta + \epsilon$$

with restrictions

$$H\beta = h, \quad (1)$$

where Y is an $n \times 1$ vector, X is an $n \times q$ matrix, h is an $\ell \times 1$ vector of known constants, and H is an $\ell \times q$ design matrix that expresses the structure of the prior information on β .

Goldberger (1964) has shown that the least squares estimator for β , combining the model and the restrictions (1), is

$$b^* = b + (X'X)^{-1}H'(H(X'X)^{-1}H')^{-1}(h - Hb), \quad (2)$$

where b is the unrestricted least squares estimator

For notational convenience, we define

$$W = (X'X)^{-1}$$

$$Z = (HW H')^{-1} \quad (3)$$

$$\delta = h - Hb, \quad (4)$$

then

$$b^* = b + W H' Z \delta. \quad (5)$$

From equation (5) we have

$$E(b^*) = \beta + WH'Z\delta$$

and

$$\text{var}(b^*) = \sigma^2(W - WH'ZHW).$$

Therefore, b^* is a biased estimator of β unless $\delta = 0$, i.e. the restrictions on β are exactly correct. But

$$\text{var}(b) - \text{var}(b^*) = \sigma^2 WH'ZHW,$$

which is non-negative definite, and so b^* has smaller variance than the ordinary least squares estimator b , whether or not b^* is biased.

Turning now to the residuals, let r be the vector of ordinary least squares residuals and for the restricted least squares residuals, let

$$\begin{aligned} r^* &= y - Xb^* \\ &= y - X(b + WH'Z(h - Hb)) \quad \text{from (5)} \\ &= y - XWX'y - XWH'Zh + XWH'ZHWX'y \\ &= [I - X(W - WH'ZHW)X']y - XWH'Zh \\ &= [I - X(W - WH'ZHW)X'](X\beta + \epsilon) - XWH'Zh \\ &= XWH'ZH\beta + [I - X(W - WH'ZHW)X']\epsilon - XWH'Zh. \end{aligned} \quad (6)$$

Hence

$$\begin{aligned} E(r^*) &= XWH'Z(H\beta - h) \neq 0 \quad \text{unless } \delta = 0, \\ \text{var}(r^*) &= \sigma^2[I - X(W - WH'ZHW)X'] \\ &= \sigma^2(I - T), \end{aligned} \quad (7)$$

where $T = X(W - WH'ZHW)X'$ is the equivalent of the hat matrix and $(I - T)$ is idempotent.

Theorem 1

$$\text{var}(r^*) - \text{var}(r) \text{ is non-negative definite.} \quad (8)$$

Proof

$$\begin{aligned} \text{var}(r^*) - \text{var}(r) &= \sigma^2(I - XWX' + XWH'ZHWX') - \sigma^2(I - XWX') \\ &\quad \text{from (7)} \\ &= \sigma^2 XWH'ZHWX', \end{aligned} \quad (9)$$

which is a non-negative definite quadratic form.

Theorem 2 The total variance of r^* is greater than the total variance of r ;

$$\text{tr}(\text{var}(r^*)) - \text{tr}(\text{var}(r)) = \sigma^2 \ell > 0. \quad (10)$$

Proof

$$\begin{aligned} \text{tr}(\text{var}(r^*)) - \text{tr}(\text{var}(r)) &= \sigma^2 \text{tr}(XWH'ZHWX') \text{ from (9)} \\ &= \sigma^2 \text{tr}(WX'XWH'ZH) \\ &= \sigma^2 \text{tr}(HWH'Z) \\ &= \sigma^2 \ell \text{ from (3).} \end{aligned}$$

Theorem 3 (Judge, Griffiths, Hill & Lee, 1980). The generalised mean square error of r^* is less than that of r provided that

$$\delta'[H(X'X)^{-1}H']\delta/\sigma^2 \leq 1 \quad (11)$$

If the condition (11) is true, then the restricted least squares residual r^* is a better estimator of the unknown e than the ordinary least squares residual r , under the generalised mean square error criterion. Note that if the restrictions are exact, i.e. $\delta = 0$, then (11) is always true.

However, r^* is biased unless $\delta = 0$, and from (8) and (10), r^* has a larger variance than r . This would suggest that it is preferable to use the least squares residuals r in the estimation of σ^2 .

The following statistic may be used for testing whether the i^{th} data point is an outlier:

$$t_i^* = \frac{r_i^*}{s_{-i}\sqrt{1 - t_{ii}}}, \quad i = 1, \dots, n, \quad (12)$$

where t_{ii} is the i^{th} diagonal element of T , and s_{-i}^2 is the least squares estimate of σ^2 from the analysis with the i^{th} observation omitted. The statistic $(t_i^*)^2$ follows a non-central F distribution with 1 and $n - q - 1$ degrees of freedom. From (6) the non-centrality is a function of the known parameters H and h , and the unknown β . If $\delta = 0$, then $(t_i^*)^2$ will have a central F distribution. These results are proved for the general case in theorems (4) and (8).

Consider any subset r_I^* of m elements of the residuals r^* indexed by the set I . Without loss of generality, we may write

$$r_I^* = (I_m \ 0)r^*.$$

Now

$$\begin{aligned} r^* &= XWH'Z(H\beta - h) + (I - T)\epsilon \quad \text{from (6)} \\ &= \mu + (I - T)\epsilon, \end{aligned}$$

where

$$\mu = XWH'Z\delta. \quad (13)$$

Therefore

$$r_I^* = (I_m \ 0)(\mu + (I - T)\epsilon). \quad (14)$$

Let $(I - T) = N$ be partitioned as

$$N = \begin{bmatrix} N_m & N_a \\ N_a' & N_{n-m} \end{bmatrix},$$

then

$$I - T_I = N_m, \quad \text{where} \quad T_I = X_I(W - WH'ZHW)X_I'. \quad (15)$$

Note that:

$$\begin{aligned} \mu'(I - T) &= (H\beta - h)'ZHWX'(I - XWX' + XWH'ZHWX') \\ &= (H\beta - h)'(ZHWX' - ZHWX'XWX' \\ &\quad + ZHWX'XWH'ZHWX') \\ &= (H\beta - h)'ZHWX' \\ &= \mu'. \end{aligned} \quad (16)$$

Theorem 4 (Khatri, 1962) If $x \sim N(0, I)$ then $x'Qx + m'x + d$ will have a non-central χ^2 distribution if and only if:

1. $Q^2 = Q$
2. $m' = m'Q$
3. $d = \frac{1}{4}m'm$.

The degrees of freedom will be $f = \text{tr}(Q)$ and the noncentrality parameter $v = d$.

Theorem 5

$$\frac{(r_I^*)'(I - T_I)^{-1}(r_I^*)}{\sigma^2} \sim \chi_m^2(\nu), \quad (17)$$

with noncentrality $\nu = \delta'ZHWX_I'(I - T_I)^{-1}X_IWH'Z\delta$.

Proof

$$(r_I^*)'N_m^{-1}(r_I^*) = (\mu' + \epsilon'N) \begin{pmatrix} I_m \\ 0 \end{pmatrix} N_m^{-1}(I_m \ 0)(\mu + N\epsilon) \text{ from (14)}$$

$$= \epsilon' NN^* N \epsilon + 2\mu' N^* N \epsilon + \mu' N^* \mu, \quad (18)$$

$$\text{where } N^* = \begin{bmatrix} N_m^{-1} & 0 \\ 0 & 0 \end{bmatrix}.$$

1. $NN^*NN^*N = NN^*NN^*N$ since $N = (I - T)$ is idempotent
 $= NN^*N$ since $N^*NN^* = N^*$.
2. $2\mu' N^*NN^*N = 2\mu' N^*NN^*N$
 $= 2\mu' N^*N.$
3. $\frac{1}{4}[2\mu' N^*N][2\mu' N^*N]' = \mu' N^*NN^*\mu$
 $= \mu' N^*\mu.$

Hence, the conditions of theorem 4 are satisfied.

$$\begin{aligned} \mu' N^* \mu &= \delta' ZHWX' \begin{pmatrix} I_m \\ 0 \end{pmatrix} N_m^{-1} (I_m \ 0) XWH' Z \delta \\ &= \delta' ZHWX'_I (I - T_I)^{-1} X_I WH' Z \delta. \end{aligned} \quad (19)$$

$$\text{tr}(NN^*N) = \text{tr}(NN^*) = m. \quad (20)$$

Theorem 6

$$\frac{(r^*)'(r^*)}{\sigma^2} \sim \chi_f^2(\nu), \quad (21)$$

with degrees of freedom $f = n - q + \ell$ and non-centrality $\nu = \delta' Z \delta$.

Proof

$$\begin{aligned} (r^*)'(r^*) &= [\mu + (I - T)\epsilon]'[\mu + (I - T)\epsilon] \text{ from (13)} \\ &= \epsilon'(I - T)\epsilon + 2\mu'(I - T)\epsilon + \mu'\mu. \end{aligned}$$

1. $(I - T)$ is idempotent.

2. $2\mu'(I-T)(I-T) = 2\mu'(I-T).$
3. $\frac{1}{4}[2\mu'(I-T)][2\mu'(I-T)]' = \mu'(I-T)\mu$
 $= \mu'\mu$ from (16).

Hence the conditions of theorem 4 are satisfied.

$$\begin{aligned}\mu'\mu &= \delta'ZHWX'XWH'Z\delta \\ &= \delta'Z\delta.\end{aligned}\tag{22}$$

$$\begin{aligned}\text{tr}(I-T) &= \text{tr}(I_n - XWX' + XWH'ZHWX') \\ &= n - \text{tr}(WX'X) + \text{tr}(WH'ZHWX'X) \\ &= n - q + \text{tr}(ZHHH') \\ &= n - q + \ell.\end{aligned}\tag{23}$$

Theorem 7

$$\frac{(S_{-I}^*)^2}{\sigma^2} \sim \chi_f^2(\nu),\tag{24}$$

with degrees of freedom $f = n - q - m + \ell$

and non-centrality $\nu = \delta'Z[I - HWX'_I(I - T_I)^{-1}X_IWH'Z]\delta$

where $(S_{-I}^*)^2 = (r^*)'(r^*) - (r_I^*)'(I - T_I)^{-1}(r_I^*)$.

Proof

$$\begin{aligned}
(S_{-I}^*)^2 &= (r^*)'(I - N^*)(r^*) \\
&= (\mu + N\epsilon)'(I - N^*)(\mu + N\epsilon) \text{ from (13)} \\
&= \epsilon'(N - NN^*N)\epsilon + 2\mu'(I - N^*)N\epsilon + \mu(I - N^*)\mu. \quad (25)
\end{aligned}$$

1.

$$\begin{aligned}
(N - NN^*N)(N - NN^*N) &= N - 2NN^*N + NN^*NNN^*N \\
&= N - NN^*N \text{ using } N^2 = N \text{ and} \\
&\quad NN^*NN^* = N^*. \quad (26)
\end{aligned}$$

2.

$$\begin{aligned}
2\mu'(I - N^*)N(N - NN^*N) &= 2\mu'(I - N^*)(N - NN^*N) \\
&= 2\mu'(N - N^*N - NN^*N + N^*NN^*N) \\
&= 2\mu'(NN - NN^*N) \text{ since } N = NN \\
&= 2\mu'(I - N^*)N \\
&= 2\mu'(I - N^*)N \text{ using (16)}.
\end{aligned}$$

3.

$$\begin{aligned}
\frac{1}{4}[2\mu'(I - N^*)N][2\mu'(I - N^*)N]' &= \mu'(I - N^*)NN(I - N^*)\mu \\
&= \mu'(N - N^*N)(N - NN^*)\mu \\
&= \mu'N(N - N^*N)(N - NN^*)N\mu \\
&\quad \text{using (16)} \\
&= \mu'(N - NN^*N)(N - NN^*N)\mu
\end{aligned}$$

$$\begin{aligned}
&= \mu'(N - NN^*N)\mu \text{ from (26)} \\
&= \mu'(NN - NN^*N)\mu \\
&= \mu'(I - N^*)\mu \text{ using (16).}
\end{aligned}$$

Hence the conditions of theorem 4 are satisfied:

$$\begin{aligned}
\mu'(I - N^*)\mu &= \mu'\mu - \mu N^*\mu \\
&= \delta'Z[I - HWX_I'(I - T_I)^{-1}X_IWH'Z]\delta \\
&\quad \text{from (22) and (19).}
\end{aligned}$$

$$\begin{aligned}
tr(N - NN^*N) &= tr(N) - tr(NN^*N) \\
&= n - q + \ell - m \text{ from (23) and (20).}
\end{aligned}$$

Theorem 8

$$(r_I^*)'(I - T_I)^{-1}(r_I^*) \text{ and } (S_{-I}^*)^2 \text{ are independently distributed.} \quad (27)$$

Proof

Using results by Searle (1971) on the independence of the distributions of quadratic forms, we need only show that the product of the matrices involved in the quadratic forms is zero.

$$\begin{aligned}
(r_I^*)'(I - T_I)^{-1}(r_I^*) &= \epsilon'NN^*N\epsilon + 2\mu'N^*N\epsilon + \mu'N^*\mu \text{ from (18).} \\
(S_{-I}^*)^2 &= \epsilon'(N - NN^*N)\epsilon + 2\mu'(I - N^*)N\epsilon + \mu'(I - N^*)\mu \text{ from (25).} \\
\text{But } NN^*N(N - NN^*N) &= NN^*N - NN^*N = 0 \text{ using } N^2 = N \text{ and } \\
N^*NN^* &= N^*.
\end{aligned}$$

Let us now focus on the least squares residuals.

Let $S_{-I}^2 = r'r - r_I'(I - V_I)^{-1}r_I$ where r are the least squares residuals, r_I is an $m \times 1$ subvector of r and V_I is the corresponding $m \times m$ submatrix of the hat matrix $V = X(X'X)^{-1}X'$.

Let $I - V = M$ be partitioned as

$$M = \begin{pmatrix} M_m & M_a \\ M'_a & M_{n-m} \end{pmatrix}.$$

Then $I - V_I = M_m$ and since $r = (I - V)\epsilon$, r_I can be written as $r_I = (I_m \ 0)(I - V)\epsilon$.

Hence

$$S^2_{-I} = \epsilon' M \epsilon - \epsilon' M \begin{pmatrix} I_m \\ 0 \end{pmatrix} M_m^{-1} (I_m \ 0) M \epsilon \quad (28)$$

$$= \epsilon' (M - M M^* M) \epsilon \text{ as } M = I - V \text{ is idempotent,} \quad (29)$$

where $M^* = \begin{pmatrix} M_m^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$

Theorem 9

$$\langle r_I^* \rangle' \langle I - T_I \rangle^{-1} \langle r_I^* \rangle \text{ and } S^2_{-I} \text{ are independently distributed.} \quad (30)$$

Proof

$$S^2_{-I} = \epsilon' (M - M M^* M) \epsilon \text{ from above.} \quad (31)$$

$$\begin{aligned} N N^* N (M - M M^* M) &= N N^* N M - N N^* N M M^* M \\ &= N N^* M - N N^* M M^* M \end{aligned}$$

since $N M = M$ by direct multiplication

$$= N N^* M - N N^* M = 0 \text{ since } N^* M M^* = N^*.$$

We are now able to generalise the test for outliers. The extension of (12) to a group of m observations is

$$(t_I^*)^2 = \frac{\langle r_I^* \rangle' \langle I - T_I \rangle^{-1} \langle r_I^* \rangle}{m s^2_{-I}}. \quad (32)$$

From (17) and (28), $(t_I^*)^2$ follows a non-central F distribution with m and $n - q - m$ degrees of freedom and non-centrality given in (17). If the restrictions on β are exact, then the distribution will be central F .

From (8) and (10) it would appear that the ordinary least squares residuals should always be used to estimate σ^2 . The estimator $s_{-I}^2 = S_{-I}^2/(n-q-m) \sim \text{central } \sigma^2\chi^2$ with $n-q-m$ degrees of freedom. Note, however, from (24) that the estimator $(s_{-I}^*)^2 = (S_{-I}^*)^2/(n-q-m+\ell) \sim \text{central } \sigma^2\chi^2$ with $n-q-m+\ell$ degrees of freedom if the restrictions on β are exact. Hence, if $\delta = 0$, we may gain ℓ degrees of freedom (where ℓ is the number of linear restrictions on β) by using the estimator $(s_{-I}^*)^2$. This may be valuable in cases where the number of degrees of freedom is small. The statistic

$$(t_I^{**})^2 = \frac{(r_I^*)'(I - T_I)^{-1}(r_I^*)}{m(s_{-I}^*)^2} \quad (33)$$

may be used for outlier testing if $H\beta = h$. From (24) and (27), $(t_I^{**})^2$ follows a central χ^2 distribution with m and $n-q-m+\ell$ degrees of freedom provided $\delta = 0$.

Thus, there may be considerable advantages by using

$(t_I^{**})^2$ rather than $(t_I^*)^2$ or even $(t_I)^2$ if the restrictions are correct (exact).

We illustrate this quite clearly in our examples for the case where $m = 1$ using the t-statistic for a single observation. From (12) where $m = 1$

$$t_i^* = \frac{r_i^*}{s_{-i}\sqrt{1-t_{ii}}}, \quad i = 1, \dots, n, \quad (34)$$

while

$$t_i^{**} = \frac{r_i^*}{s_{-i}^*\sqrt{1-t_{ii}}}, \quad i = 1, \dots, n, \quad (35)$$

and

$$t_i = \frac{r_i}{s_{-i}\sqrt{1-h_{ii}}}, \quad i = 1, \dots, n, \quad (36)$$

with h_{ii} the i^{th} diagonal of the hat matrix $X(X'X)^{-1}X'$.

If the restrictions are exact the statistic t_i^* will follow a Student's t -distribution with $n-q-1$ degrees of freedom. As test statistic for a single outlier we will then use $\max |t_i^*|$ where percentage points for this statistic

can be found from Weisberg's (1980) tables. For the statistic t_i^{**} the degrees of freedom will be $n - q - 1 + \ell$.

If $\delta \neq 0$, then $(t_I^{**})^2$ is doubly non-central, and the non-centrality is unknown. In this case, it is recommended that the statistic $(t_I^*)^2$ be used. Let us now consider a measure of the influence of a group of m observations indexed by the set I . The influence of a single observation is simply the special case $m = 1$. An appropriate measure would be

$$\Delta_I^* = \frac{(b_{-I}^* - b^*)' X' X (b_{-I}^* - b^*)}{qs_{-I}^2}. \quad (37)$$

Cook's distance D_I^* is given by

$$D_I^* = \frac{(b_{-I}^* - b^*)' X' X (b_{-I}^* - b^*)}{qs^2}, \quad (38)$$

where s^2 is the least squares estimate of σ^2 . Following Belsley, Kuh and Welsch (1980) we prefer Δ_I^* as a measure of influence. This measure, in a modified form, is also preferred by Atkinson (1981).

There are considerable advantages from using Δ_I^* rather than Cook's distance D_I^* . Exact distributional results can be derived for Δ_I^* , and when the restrictions are exact, the distributions are weighted sums of central F-distributions. As far as our knowledge goes, the exact distributional result for Cook's distance is unknown. These results are discussed in the next paragraph.

The $100(1-\alpha)\%$ confidence ellipsoid for the unknown parameter β is given by the set of all vectors b that satisfy

$$\frac{(b - \beta)' X' X (b - \beta)}{qs^2} \leq F_{1-\alpha}(q, n - q),$$

where $F_{1-\alpha}(q, n - q)$ is the upper α point of the F distribution with q and $n - q$ degrees of freedom.

It is readily seen that D_i^* and Δ_i^* provide a measure of the distance between b_{-i}^* and b in terms of significance levels. For example, if D_i^* is near to $F_{.4}(q, n - q)$ then the removal of the i^{th} observation has the effect of moving the estimate for β to the edge of the 40% confidence ellipsoid for β based on the full data set, which may be cause for some concern. Ideally, we would like all the b_{-i}^* to remain within say a 5% confidence region. A cut-off value for D_i^* would then be the lower (5 or 10) percentage points of the F-distribution with q and $n - q$ degrees of freedom. For Δ_i^* the degrees of freedom will be q and $n - q - 1$. From (5), the nature of the estimator b^* does not allow a simple expression for $(b_{-I}^* - b^*)$. From (5)

$$b^* = (I - WH'ZH)b + WH'Zh. \quad (39)$$

$$\text{Let } W_{-I} = (X'_{-I}X_{-I})^{-1} \quad (40)$$

$$Z_{-I} = (HW_{-I}H')^{-1} \quad (41)$$

$$\text{then } b_{-I}^* = (I - W_{-I}H'Z_{-I}H)b_{-I} + W_{-I}H'Z_{-I}h. \quad (42)$$

$$\begin{aligned} \text{Now } H(b_{-I}^* - b^*) &= Hb_{-I} - HW_{-I}H'Z_{-I}Hb_{-I} + HW_{-I}H'Z_{-I}h \\ &\quad - Hb + HWH'ZHb - HWH'Zh \end{aligned}$$

from (39) and (42)

$$= Hb_{-I} - Hb_{-I} + h - Hb + Hb - h$$

using (3) and (41)

$$= 0. \quad (43)$$

The equation (43) is a system of ℓ simultaneous equations in q unknowns. An analytic solution for $(b_{-I}^* - b^*)$ will exist only if the rank of H is $\geq q$; that is, the number of constraints must be at least as great as the number of variables

in the regression. In practice, the number of constraints ℓ will tend to be small, so we shall not be able to find $(b_{-I}^* - b^*)$ in general.

We would nevertheless like a computational expression for b_{-I}^* which would enable us to compute its value without the need for performing a complete regression on the truncated data set.

Bingham (1977) showed that

$$W_{-I} = W + W X_I' (I - V_I)^{-1} X_I W, \quad (44)$$

where $V_I = X_I W X_I'$ as before.

Using (44) we have

$$Z_{-I} = (H W H' + H W X_I' (I - V_I)^{-1} X_I W H')^{-1}. \quad (45)$$

$$\begin{aligned} \text{Now } b_{-I}^* &= (I - W_{-I} H' Z_{-I} H) b_{-I} + W_{-I} H' Z_{-I} h \text{ from (42)} \\ &= (I - W_{-I} H' Z_{-I} H) (b - W X_I' (I - V_I)^{-1} r_I) + W_{-I} H' Z_{-I} h \\ &= [I - (W + W X_I' (I - V_I)^{-1} X_I W) H' Z_{-I} H] (b - W X_I' (I - V_I)^{-1} r_I) \\ &\quad + (W + W X_I' (I - V_I)^{-1} X_I W) H' Z_{-I} h \text{ from (44)} \\ &= [I - (I + W X_I' (I - V_I)^{-1} X_I) W H' Z_{-I} H] b \\ &\quad - [I - (I + W X_I' (I - V_I)^{-1} X_I) W H' Z_{-I} H] W X_I' (I - V_I)^{-1} r_I \\ &\quad + (1 + W X_I' (I - V_I)^{-1} X_I) W H' Z_{-I} h \\ &= (I - C H) b - (I - C H) W X_I' (I - V_I)^{-1} r_I + C h \\ &= b + C(h - H b) - (I - C H) W X_I' (I - V_I)^{-1} r_I, \end{aligned} \quad (46)$$

$$\text{where } C = (I + W X_I' (I - V_I)^{-1} X_I) W H' Z_{-I}. \quad (47)$$

Hence the procedure for computing b_{-I}^* would be

1. compute the value of $Z_{-I}(\ell \times \ell)$ from (41)

2. compute the value of $C(q \times \ell)$ from (47)
3. compute b_{-I}^* from (42).

The procedure seems clumsy, with many matrix multiplications and inversion of the matrices $(I - V_I)$ and $(HWH' + HWX_I'(I - V_I)^{-1}X_IWH')$. However, these matrices are $m \times m$ and $\ell \times \ell$ respectively and hence small, and in fact many sequences of multiplication of small matrices are repeated and need to be computed once only.

3. Examples of RLS Estimators

The subset selection problem in multiple regression can be described as that of minimizing the residual sums of squares for the full model, subject to the restriction that certain of the coefficients are zero. That is minimize

$$Q(\beta) = (Y - X\beta)'(Y - X\beta)$$

subject to $\beta_i = 0, i \in R$ where R is a particular set of indices.

The ridge estimator of Hoerl and Kennard (1970) is the solution to the problem: Minimize

$$(\beta - \hat{\beta})'X'X(\beta - \hat{\beta}),$$

subject to $\beta'\beta \leq d^2$ where the radius of d depends on the ridge constant k .

The shrinkage estimator by Stein (1960) shrinks the least squares estimator towards the origin. It is also a restricted estimator

If $\tilde{\beta} = c\hat{\beta}$, then $\tilde{\beta}$ is the solution to the problem: Minimize

$$(\beta - \hat{\beta})'(\beta - \hat{\beta})$$

subject to $\beta'\beta \leq d$, the radius d being determined from the constant c .

The principal component estimator is also a restricted least squares estimator. Let Λ denote the eigenvalues of $X'X$, and P the orthogonal matrix of eigenvectors. That is $P'X'XP = \Lambda$ with $P'P = I$. Letting $Z = XP$, $\alpha = P'\beta$, then the least squares model (1) becomes

$$Y = Z\alpha + \epsilon,$$

with the least squares estimate determined by

$$\Lambda\hat{\alpha} = Z'Y,$$

and

$$\hat{\beta} = P\hat{\alpha}.$$

Let us order the eigenvalues and eigenvectors so that Z_i corresponds to the i^{th} smallest root. Now if s of the eigenvalues are zero and if we write

$$P = (P_r \ P_s)$$

and

$$\Lambda = \begin{pmatrix} \Lambda_r & 0 \\ 0 & \Lambda_s \end{pmatrix},$$

where $r + s = q$, then the principal component estimator is $\bar{\alpha}_r = \Lambda_r^{-1} P_r' X' Y$ and

$$\bar{\beta}_r = P_r \bar{\alpha}_r.$$

Thus the last s eigenvalues are eliminated.

This is also equivalent to the solution of the problem: Minimize

$$(\beta - \hat{\beta})' X' X (\beta - \hat{\beta}) \text{ subject to } P_s' \beta = 0.$$

Troskie and Chalton (1998) have derived the exact distribution of

$$\Delta_I^* = \frac{(b_{-I}^* - b^*)' X' X (b_{-I}^* - b^*)}{qs_{-I}^2}, \quad (48)$$

when $b_{-I}^* = b_{PC}$, that is the principal component estimator is used with exact restrictions. They have shown that the distribution of Δ_I^* is that of a weighted sum of F-distributions. If m , the number of potential outliers, is small

($m = 2$ or 3) then approximate upper percentage points may be derived for the distribution of Δ_T^* .

The fractional rank and Sclove estimators are also RLS estimators (see Hocking (1976)).

4. A Simulation Study

The effectiveness of the techniques discussed in this paper may be tested using a contrived or simulated data set. In such a data set, all parameters are known, and the extent of collinearity between the variables can be controlled.

In this section, we shall examine in detail how the various techniques perform in the presence of known irregularities.

A data set of 40 observations and 5 independent variables was generated as described below. The first independent variable X_1 is a trend variable, typical of many time series data sets: $X_1 = i$ for observation $i, i = 1, \dots, n$. The variables X_2, X_3 and X_4 are independently normally distributed, with $X_2 \sim N(5, 1)$ and $X_3, X_4 \sim N(10, 1)$. An approximate linear relationship between the independent variables was introduced by making $X_5 = X_3 + X_4 + e$ where $e \sim N(0, 0.1^2)$. Hence the linear relationship is not exact, but the correlation between X_5 and $(X_3 + X_4)$ is $> .999$. The dependent variable was generated as

$$Y = 100 + 2X_1 + 10X_2 + 10X_3 + 10X_4 + 10X_5 + \epsilon,$$

where $\epsilon \sim N(0, 5^2)$.

The following artificial extreme cases were introduced into the data set: Firstly, in observation 1, the variable X_2 was given a value of 10 (X_2 is usually centred on 5); also in generating Y , the random error term e was made to be

zero. Hence, observation 1 has high leverage in an independent variable not affected by the collinearity, and the observation is not an outlier.

Secondly, in observations 2 and 3, X_3 and X_4 were centred on 15 instead of 10, and corresponding adjustments were made in the values of X_5 and Y . The result is that these observations have very high values of Y , X_3 , X_4 and X_5 . In addition, a possible outlier was introduced by subtracting 10 from the Y value of observation 2 only.

Thirdly, observation 4 is made to be a straightforward outlier by adding 20 to its Y value.

The complete data set and a summary of the rules used to generate it can be obtained from the authors.

The analysis initially warns us of the severe collinearity problem ; a condition number of 3962 and smallest root of 0.0007. For restrictions using ridge we use a ridge constant $k = 0.1$ and for restrictions using principal components we use $s = q$ with $r = q - 1$, that is only the last root is eliminated so that the exact restriction is $P'_q\beta = 0$, that is P_q is the last vector and the last root λ_q is eliminated. Observation 4 is identified as a significant outlier with the studentised residual $t_4 = 3.836$, the critical value of $\max |t_i|$ being 3.53 at the 5 % significance level from Weisberg's Tables. The leverage of observation 4 (0.358) is not unusual, yet it is still the most influential observation in the sample, with $\Delta_4 = 0.879$. The 5% lower critical point of the F-distribution for $q=6$ and $40-6-1=33$ degrees of freedom is 0.4237. The value of $\Delta_4 = 0.879$ is even larger than the 25% lower percentage point of the F-distribution which is 0.7246. This gives sufficient evidence to eliminate observation 4 as an outlier of significant influence.

When observation 4 has been removed, further analysis reveals that there are no significant outliers in the sample. The highest value of t_i corresponds to observation 34 ($t_{34} = -3.14$), but it is not significant even at the 10% level; nor does this observation have any significant influence. Observations with high leverage are numbers 1, 2 and 3 (as we would hope) with leverage 0.99, 0.69 and 0.71 respectively. The influence measures are $\Delta_1 = 0.055$, $\Delta_2 = 0.991$ and $\Delta_3 = 0.713$. Observations 2 and 3 are by far the most influential, corresponding to 45.5% and 27.9% confidence ellipsoids respectively. Yet jointly observations 2 and 3 are uninfluential with $\Delta_{2,3} = 0.134$ only. This is apparently because one residual is positive and the other negative.

A ridge trace can be performed on any statistic or diagnostic by plotting the statistic or diagnostic against the ridge constant k . This is particularly useful when collinearities are present in the data. Let us examine the ridge trace for the influence of selected groups of size 2 (Figure 1). It is seen that for large k , the group of 2,3 dominates.

When groups of size 3 are examined, the pair 2,3 does not feature in the ordinary least squares analysis. However, when the ridge trace is applied, this pair dominates 8 of the ten most influential groups.

There is sufficient evidence that the analyst should pay particular attention to observations 2 and 3, which are individually influential, and also as a pair, although this latter fact is obscured by the collinearity in the data, and emerges only in the ridge trace. However, the least squares estimate of the regression coefficients would not change radically if this pair were to be removed, because $\Delta_{2,3}$ is small. The ridge estimate will change, as the pair becomes influential for larger values of k .

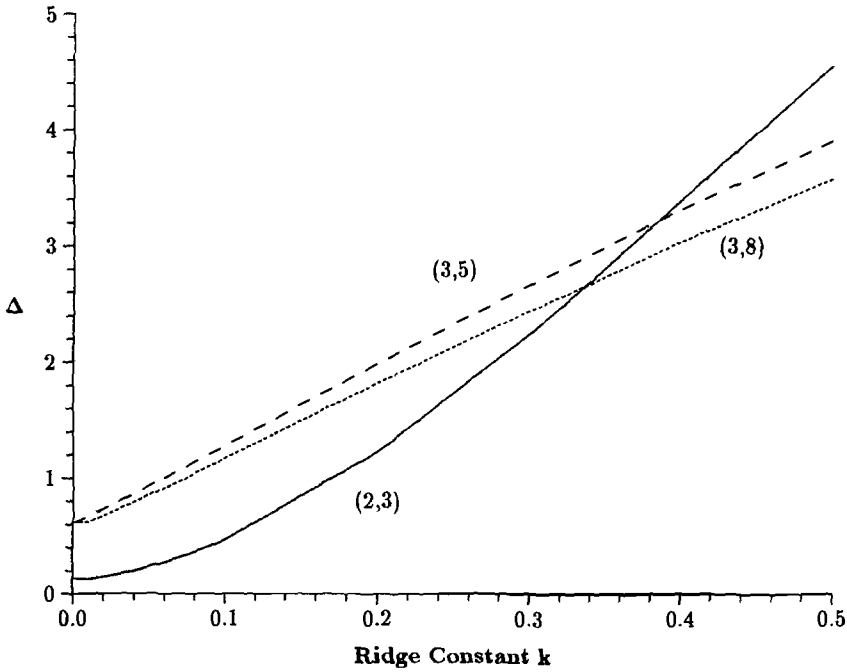


Figure 1. Ridge trace for the influence of selected groups

Whether or not the pair 2,3 is removed, all techniques (ordinary least squares, ridge and PC regression) recognize observation 1 as a point of high leverage without significant influence. There is nothing to be gained from deleting the observation. If this were a real analysis, the researcher should note the presence of the high leverage point, and possibly investigate why such an observation is so different from the rest of the data set, yet consistent with the model.

Table 1 below lists the estimates of the regression parameters at various stages of the analysis.

Table 1. Regression parameters in simulation study

	True value	Ordinary least squares	Ridge k = 0.1	PC-estimate
<u>Complete data set:</u>				
Constant	100	113.47	145.36	116.45
x_1	2	2.04	1.76	2.01
x_2	10	9.93	8.04	9.94
x_3	10	22.88	10.58	11.36
x_4	10	24.03	12.52	12.23
x_5	10	-4.08	7.05	7.47
<u>Outlier 4 deleted:</u>				
Constant	100	111.02	139.86	110.79
x_1	2	2.07	1.81	2.08
x_2	10	9.60	7.87	9.60
x_3	10	10.68	10.83	11.78
x_4	10	10.84	12.21	11.97
x_5	10	9.76	7.30	7.65
<u>2, 3, 4 deleted</u>				
Constant	100	108.84	126.62	108.98
x_1	2	2.04	1.79	2.03
x_2	10	9.23	7.62	9.22
x_3	10	11.67	9.91	10.31
x_4	10	12.77	11.69	11.37
x_5	10	7.54	8.75	8.92

The only really significant impact on the parameter estimation was due to the removal of the outlier observation 4; prior to this, the least squares estimate was particularly poor. When restrictions are imposed using ridge or principal components the estimates were considerably better and in fact similar. These two techniques appear to be more robust than least squares. The performance of the principal component estimator is especially satisfying. With the roots

very small the restrictions can be considered to be exact. All the distributional results will therefore be central. Computationally this estimator is far easier to compute than the ridge estimator and also avoids the problems of estimating the ridge constant k . There may well be improvements by using the fractional rank estimator, but this may be marginal.

In Figure 2 we plot the studentised residuals t_i for the OLS ($k = 0.0$), and t_i^* for ridge residuals ($k = 0.1$), and for the principal component estimator.

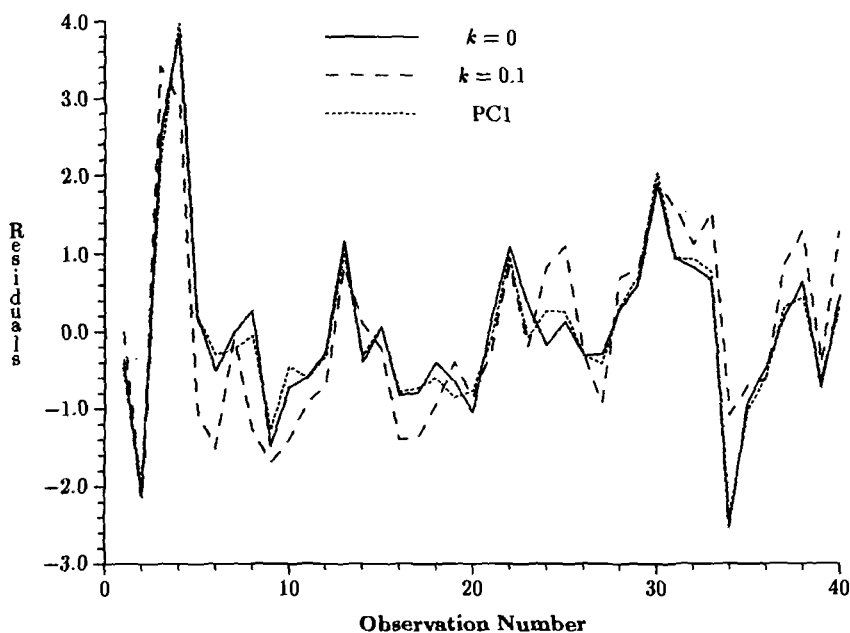


Figure 2. Studentised residuals

The two smallest eigenvalues are 0.3784 and 0.0007. We thus assume $s = q, r = q-1$ so that the restriction is $P'_q\beta = 0$ and the smallest eigenvalue is eliminated. Since the smallest eigenvalue is very small we can assume that the restriction is exact and the t_i have central t distributions. All three techniques identify observation 4 as having the largest studentised residual with values of 3.84^* ($k = 0.0$), 2.92 ($k = 0.1$) and 3.98^* ($PC1$) (* significant at 5% level).

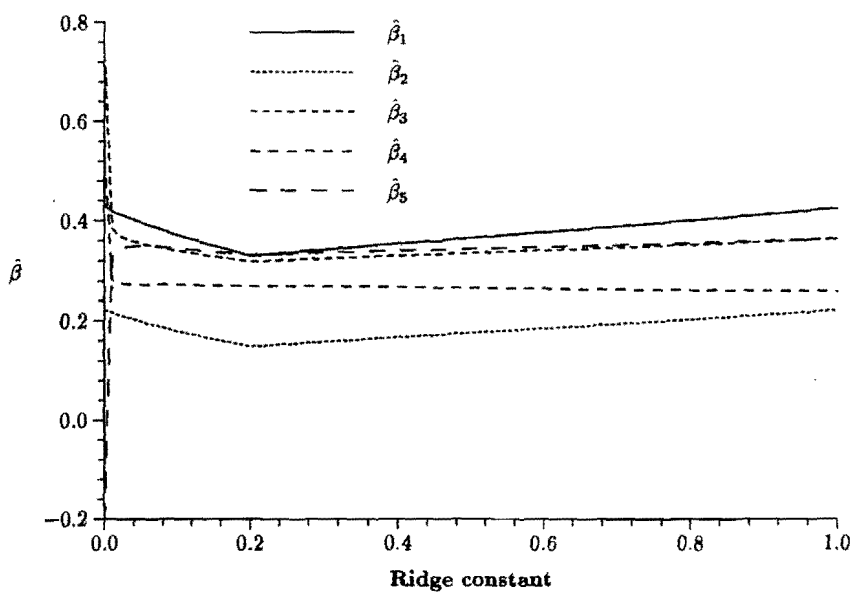
Lack of accuracy in the parameter estimation is probably due to the severe collinearity in the data set. Recognizing that the collinearity is due to the relationship between X_5 and other variables, suppose we effectively remove X_5 by applying restricted least squares regression with the constraint $\beta_5 = 0$. Since the true values of β_3, β_4 and β_5 are 10, and $X_5 = X_3 + X_4$, the new values of β_3 and β_4 are approximately 20 if β_5 is constrained to be 0.

If the variable X_5 is removed, the condition number of the new matrix $X'X$ is 4.4, and the collinearity problem is solved.

The restricted least squares analysis proceeds along similar lines to that described thus far, and all the abnormalities in the data set are successfully identified. Estimates of the regression parameters are listed in Table 2 below. Estimation is generally far more accurate than in the analysis of the unrestricted data set. Although we are able to develop workable techniques to overcome collinearity problems (ridge and PC regression), it is obvious that the analysis is far simpler and more effective if the sources of collinearity can be eliminated, in this case by using exact restrictions on the model.

Table 2. Restricted least squares estimates of regression parameters

	True value	Complete data set	Outlier 4 deleted	Obsn. 2, 3, 4 deleted
Constant	100	114.49	109.27	108.27
x_1	2	2.03	2.09	2.05
x_2	10	9.94	9.62	9.26
x_3	20	18.81	19.37	19.10
x_4	20	19.86	19.80	20.43
x_5	0	0	0	0

**Figure 3.** Ridge trace of $\hat{\beta}$'s with no restrictions

In Figure 3 we give the ridge trace of the standardised regression coefficients with no restrictions and in Figure 4 we give the ridge trace of the same regression coefficients with the restriction that $\beta_5 = 0$. It is clear that without restrictions the least squares estimates are very unstable and only stabilises around a ridge constant of $k = 0.2$. With restrictions the estimates stabilise immediately.

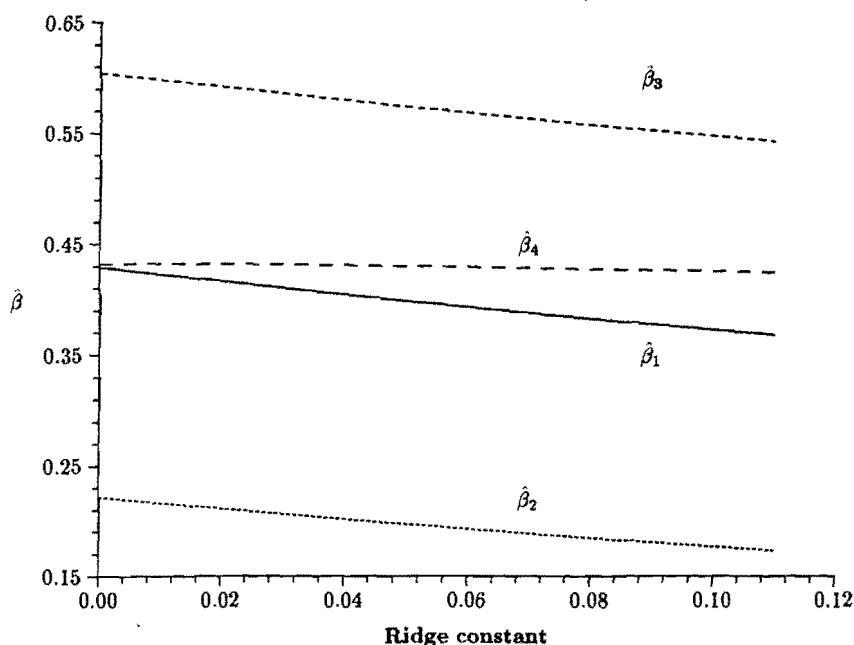


Figure 4. Ridge trace of $\hat{\beta}$'s with restriction $\beta_5 = 0$

5. Practical Example: The Consumption Function Data

The consumption function data have been analyzed by Belsley, Kuh and Welsch (1980). The data set consists of a response variable (total consumption) and four explanatory x variables.

The function is

$$C(T) = \beta_0 + \beta_1 C(T-1) + \beta_2 DPI(T) + \beta_3 r(T) + \beta_4 \Delta DPI(T),$$

with $T = 1, \dots, 27$, where C is total consumption, 1958 Dollars, DPI is the disposable personal income, 1958 dollars, r is the interest rate (Moody's Aaa) and all series are annual, 1948 – 1974. For a detailed analysis – see Belsley *et al.*

The restrictions imposed in the model are

$$H = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & .9 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} h = \begin{pmatrix} 0.7 \\ 0.9 \\ 0.25 \end{pmatrix}.$$

In addition to the restrictions Belsley *et al.* assume some prior distribution, that is, the restrictions are of the form

$$H\beta + u = h,$$

where h is an $\ell \times 1$ vector of known constants, H is an $\ell \times q$ matrix and u is an $\ell \times 1$ normally distributed random variable with mean δ and covariance matrix Σ_2 , Σ_2 known. Thus the restrictions are not exact.

In Table 3 we give the estimates for the β 's for least squares (OLS), the restricted least squares (RLS) estimator for the restrictions imposed by Belsley *et al.* and restrictions imposed by ridge using $k = 0.05$ and $k = 0.1$. For principal component regression, we note that the condition number is very large, being 8008 for the standardised $X'X$ matrix, thus indicating severe collinearity.

The two smallest eigenvalues are 0.0282 and 0.0004 and we therefore assume $s = 2$ and the restriction is $P_2'\beta = 0$, which means the two smallest roots are eliminated. Note that the estimates for ridge and PC are very similar but quite different from that of OLS and the RLS estimate of Belsley *et al.*

Table 3. Regression estimates for Belsley *et al.* data

	k = 0.0	Belsley restrictions	k = 0.05	k = 0.10	PC (2 roots eliminated)
$\hat{\beta}_0$	6.7241	7.61	22.9499	31.0569	25.6259
$\hat{\beta}_1$	0.2454	0.241	0.3984	0.3666	0.3357
$\hat{\beta}_2$	0.6984	0.688	0.3416	0.3133	0.2857
$\hat{\beta}_3$	-2.2097	-1.354	11.1179	13.8171	19.2004
$\hat{\beta}_4$	0.1608	0.198	0.6717	0.7452	0.8366

The condition number of the standardised matrix is 8008, showing severe collinearity. It is clear that only the ridge and PC procedures overcome the severe collinearity problem.

For $k = 0.0$, the highest studentised residuals are -2.44, 1.87 and 1.87 (obs. 26, 22 and 24). The $\max |t_i|$ of 2.44 is not significant since $t(\alpha/n, n - q - 1)$ for $\alpha = 0.05$, $n = 27$ and $q = 5$ from Weisberg's (1980) tables is given by 3.67. The highest studentised residuals for the Belsley *et al.* restrictions are the same three with values -2.518, 1.861 and 1.823. The $\max |t_i|$ is 2.518 and $t(\alpha/n, n + \ell - q - 1)$ for $\ell = 3$ is 3.58. Thus this observation is clearly not significant. For ridge restrictions the $\max |t_i|$ is for observation 23 with a value of 5.738 for $k = 0.05$ and 6.135 for $k = 0.10$. Given the smallest eigenvalues are 0.02822 and 0.000386, we can certainly assume that $s = 2$ and $P_2'\beta = 0$. The $\max |t_i|$ is again for observation 23 with a value of 9.119. The critical value for $t(\alpha/n, n - q - 1)$ is 3.67. Clearly, one can assume that the

restrictions are exact so that t_i follows a central t-distribution. Thus observation 23 can clearly be declared a significant outlier

6. Graphical Plots

In this section for outliers we plot the studentised residuals t_i together with the studentised residuals $t_{i(R)}$ and $t_{i(PC)}$ obtained from the restrictions on the parameters for two well-known data sets. For influential observations we plot the modified Cook's distance A_i as proposed by Atkinson(1985) where

$$A_i = \sqrt{n - q} | D_i |$$

which according to Atkinson is more amenable to graphical plots.

The first data set (Belsley, *et al.* 1980,) has a condition number of 8008 indicating severe collinearity. The second data set (Chatterjee and Hadi, 1986) has a condition number of 36 indicating no collinearity.

For the ridge estimate we use a value for both plots of $k = 0.1$. From a plot of the ridge trace of the coefficients for the Belsley *et al.* data the estimates stabilize around $k = 0.1$. For the principal component estimates the last two eigenvalues of sizes 0.0282 and 0.0004 are close enough to zero and we use a PC2 estimator with two eigenvalues eliminated. For comparisons we also use a PC2 estimator with two eigenvalues eliminated for the Chatterjee and Hadi (1986) data.

The differences in the plots are quite alarming. For the first data set, the plots of the t_i , Figure 5, and A_i , Figure 6, show marked differences between the estimators. For the second data set, the plots of the t_i , Figure 7, and A_i , Figure 8, show very similar results for the different estimators.

The effects of the presence or non-presence of collinearities are clearly demonstrated. The use of restrictions on the parameters to overcome the

harmful effect of collinearity is clearly demonstrated. In the presence of collinearity no analysis is meaningful without a correction for collinearity by means of restrictions on the parameters.

We also give a normal probability plot of the differences between the OLS and PC statistics for t_i and A_i . Again the differences in the plots are quite illuminating. The probability plots for the Belsley data, Figures 9 and 10, show heavy tails indicating large differences between the OLS and PC statistics.

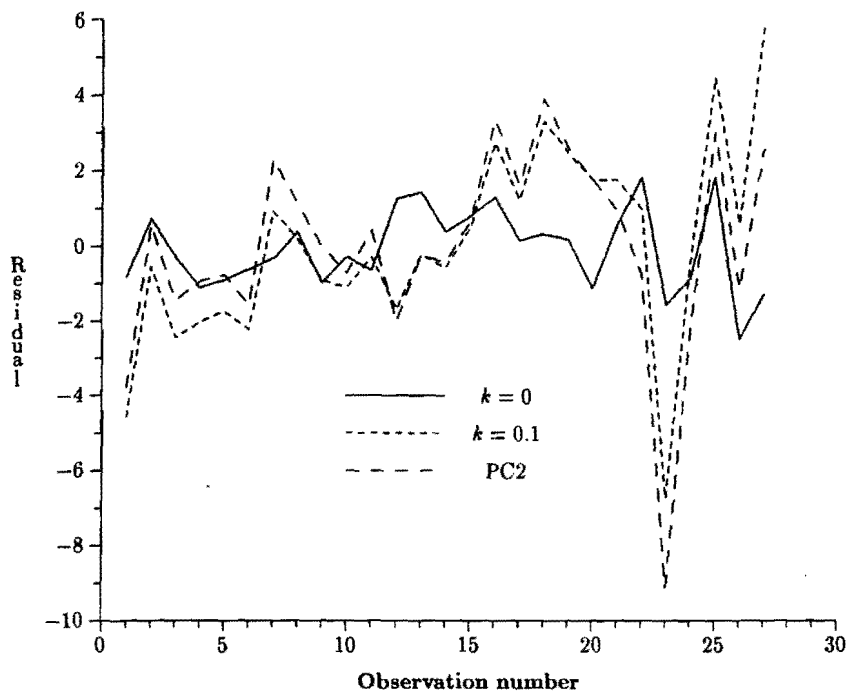


Figure 5. Studentised residuals for Belsley data

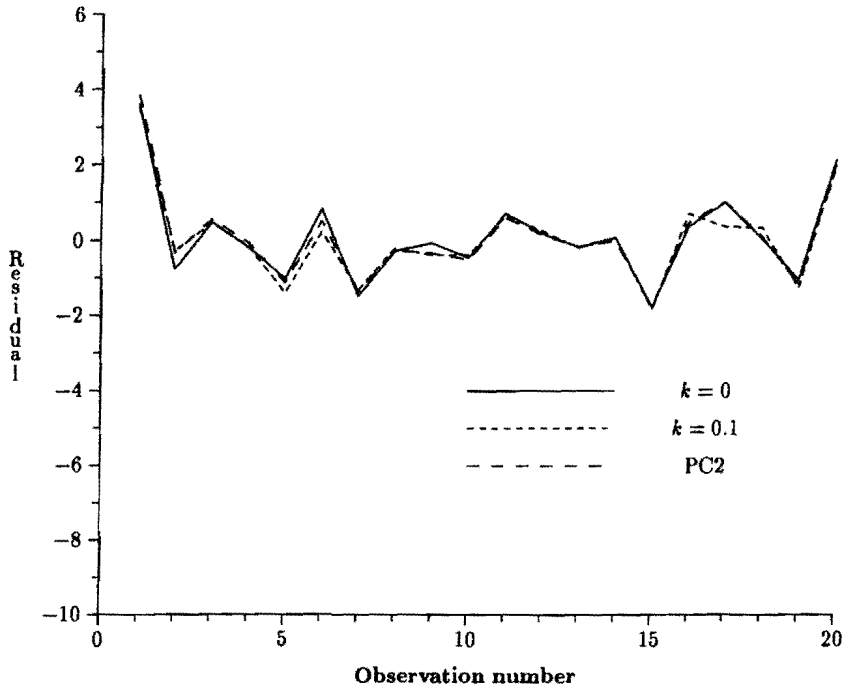


Figure 6. Studentised residuals for Chatterjee data

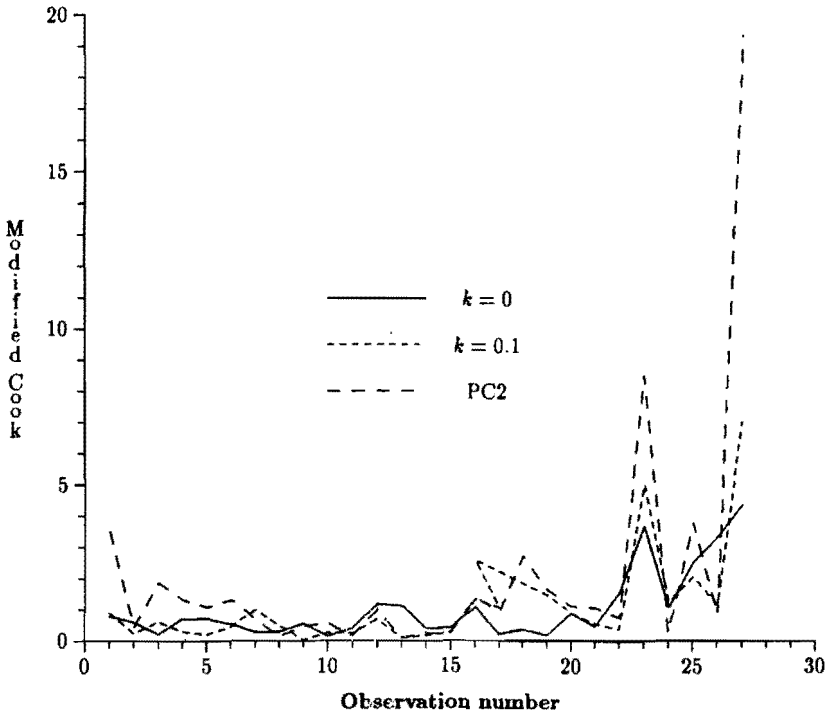


Figure 7. Modified Cook's statistics for Belsley data

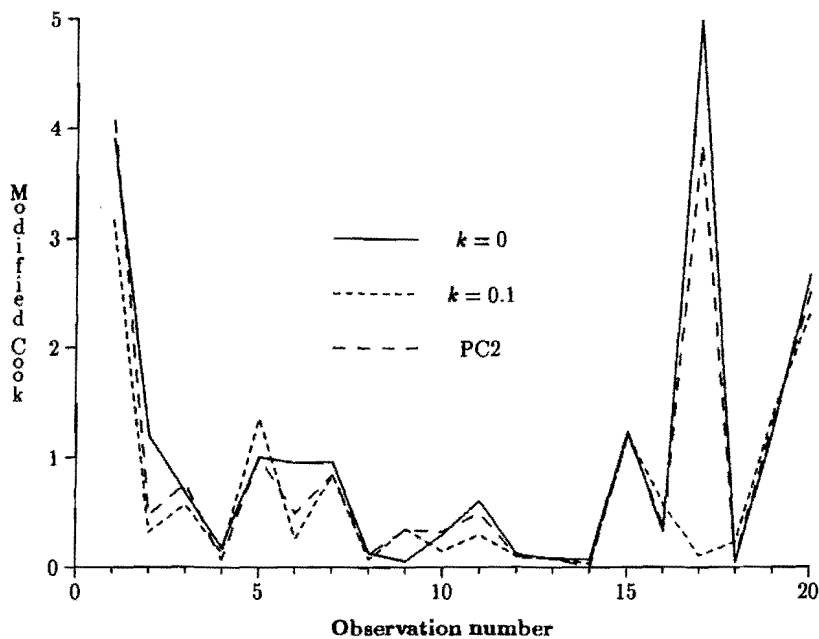


Figure 8. Modified Cook's statistics for Chatterjee data

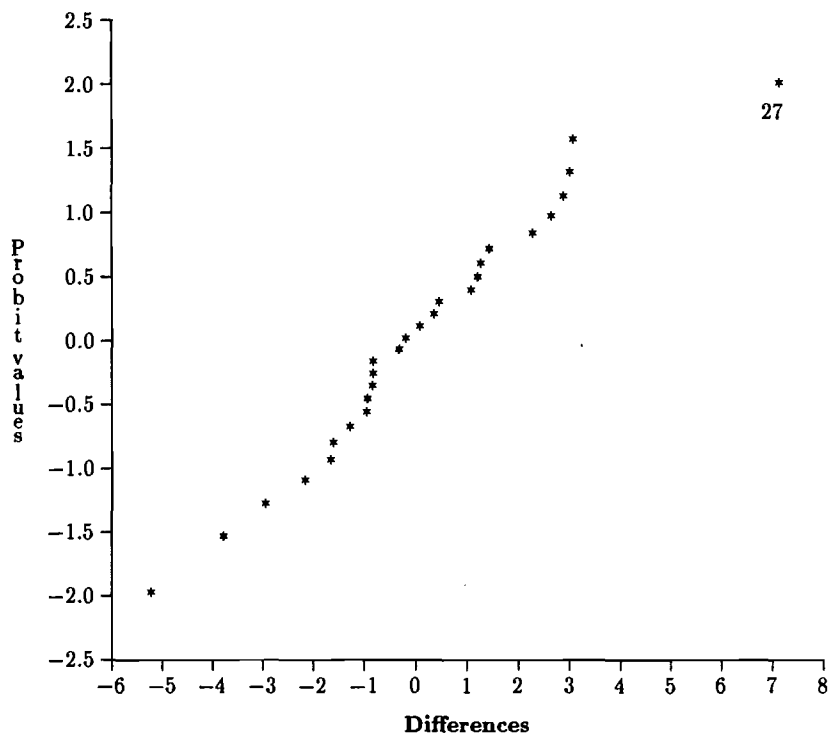


Figure 9. Normal probability plot: Belsley studentised residual differences

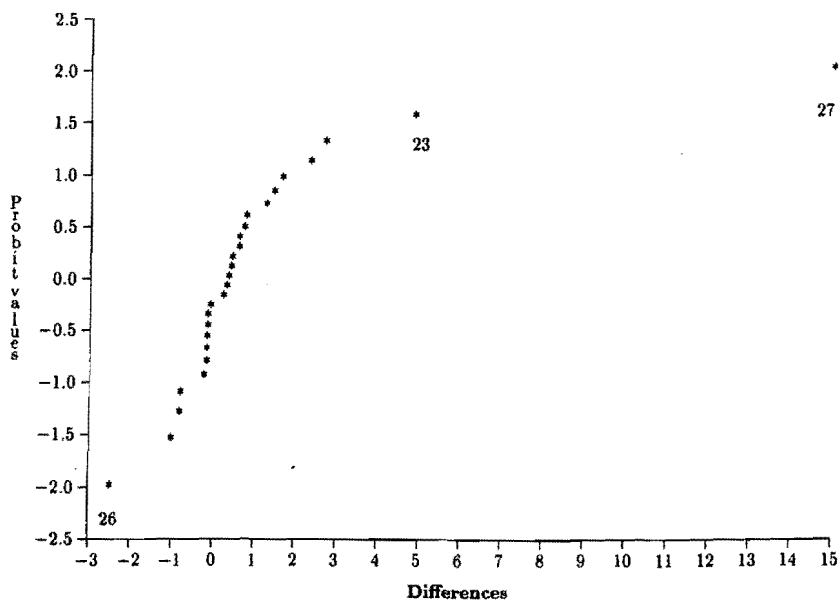


Figure 10. Normal probability plot: Belsley modified Cook's differences

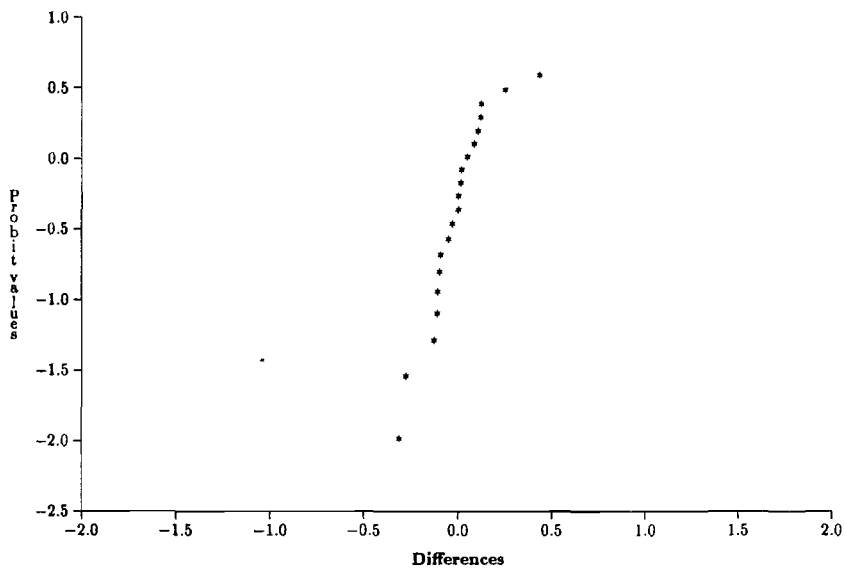


Figure 11. Normal probability plot: Chatterjee studentised residuals differences

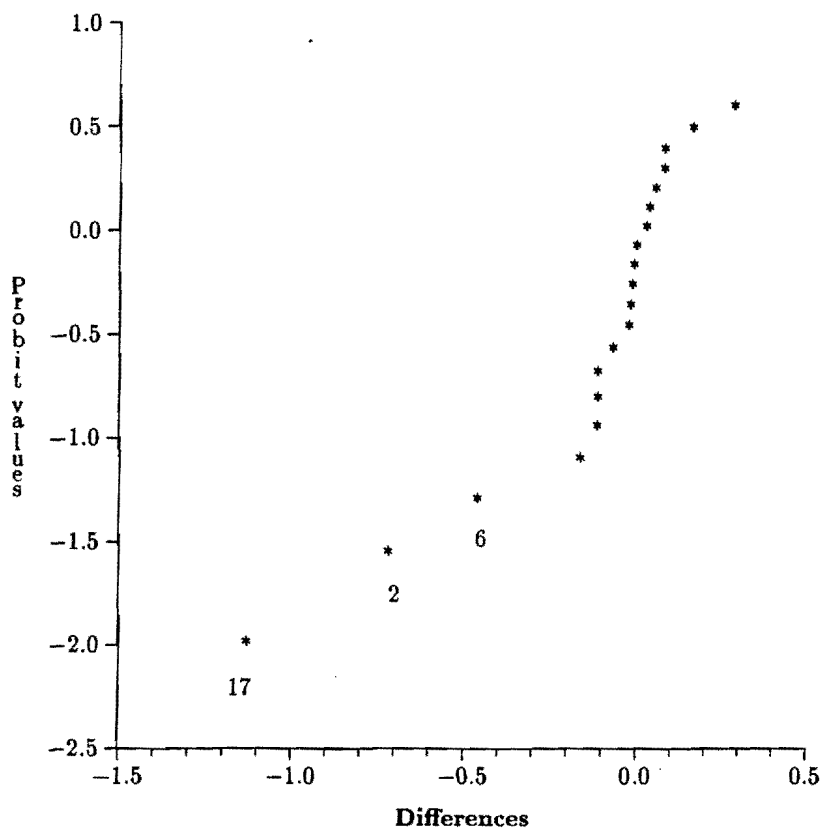


Figure 12. Normal probability plot: Chatterjee modified Cook's differences

Summary and Conclusions

In this paper, use is made of restricted least squares to estimate the regression parameters in the linear model. These estimates are then used to compute estimates of the residuals and their variances. Test statistics for outliers are derived and diagnostics to detect influential observations are given. In the latter case the computational procedures become very cumbersome. A simulated study is used which clearly demonstrates the superiority of the restricted least squares estimates. Two well-known practical examples are used, with restrictions on the parameters, which also brings new insights. The performance of the principal component estimator, as shown in both the simulation study and practical examples, is very satisfactory. It is recommended that this estimator should be considered as a very serious contender.

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